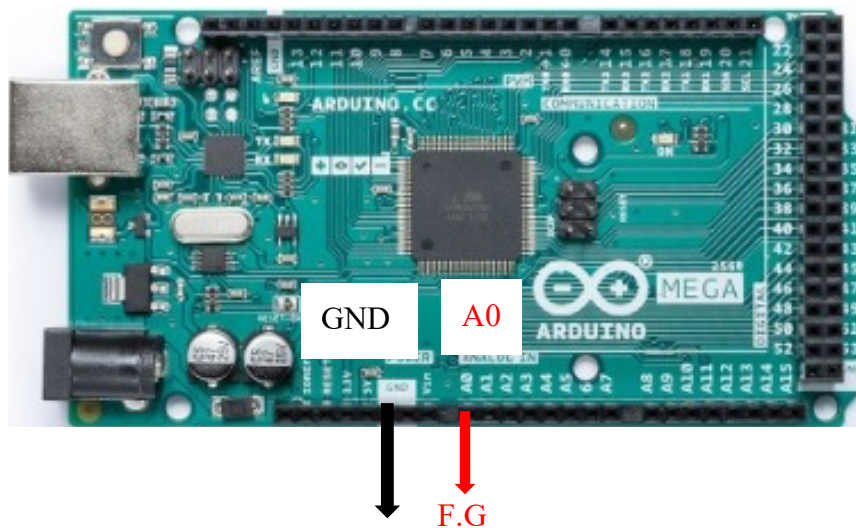


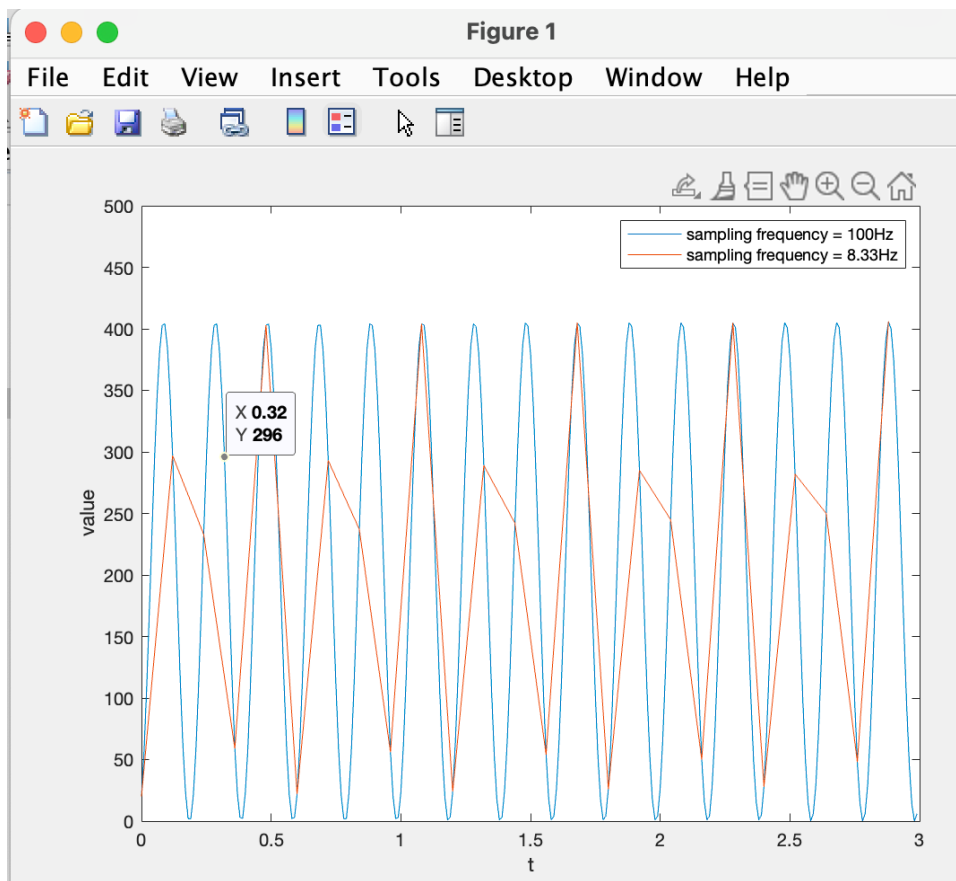
REPORT

Experiment 1: Sampling and aliasing

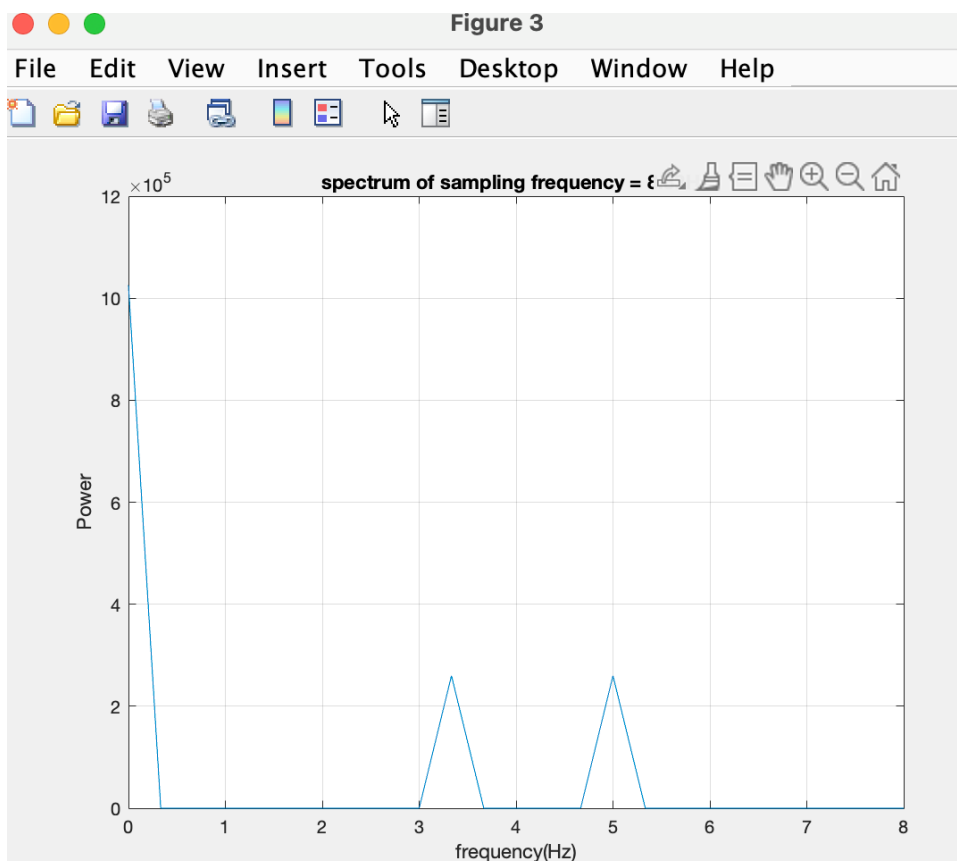
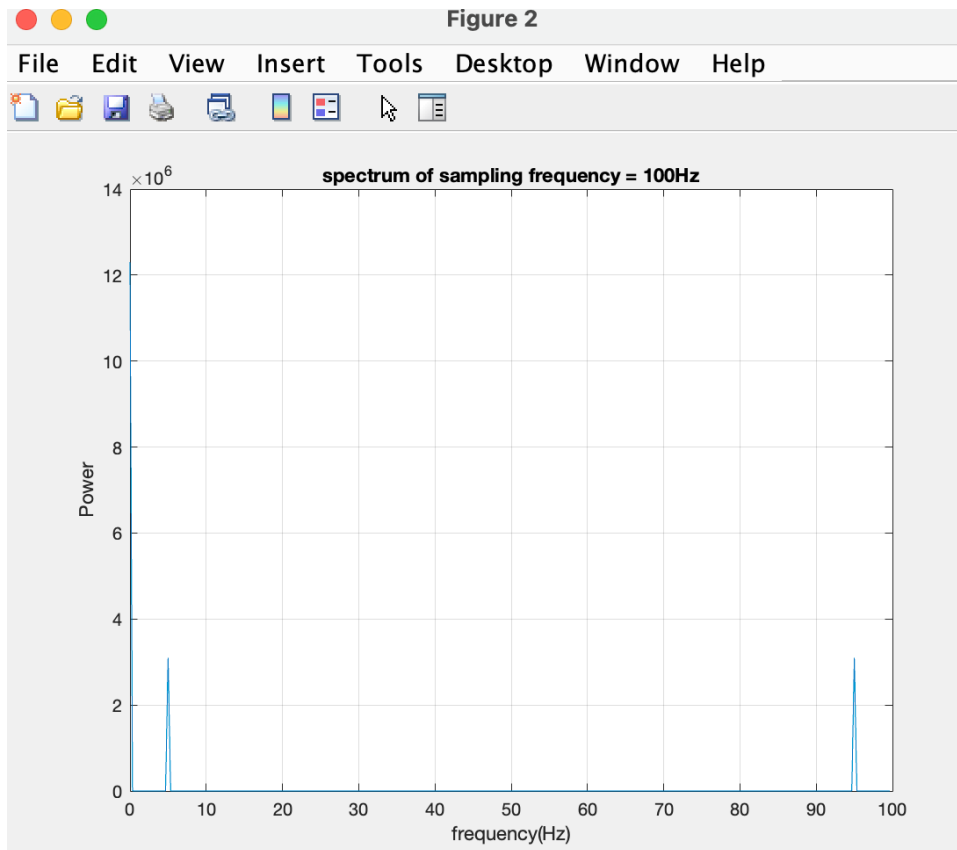


	Normal	Aliasing
Wave frequency	5Hz	3.33Hz
Sampling Frequency	100Hz	8.33Hz

1. Result of the code plot(1pic):



2. Result of the FFT plot(2pic):



Discuss:

Try to explain if aliasing happened, What's bothering you?

if sampling frequency is 8.33Hz, then since nyquist frequency= $0.5 \times \text{sampling frequency}$, $8.33/2=4.165\text{Hz}$ is lower than the wave frequency, so it will cause aliasing,(which means we would not get the real wave form by using the sampling data) and the false frequency will be sampling frequency-wave frequency= $8.33-5=3.33\text{Hz}$ (false frequency).

if we choose sampling frequency to be 100Hz, then nyquist frequency will be 50Hz and it is greater than wave frequency, thus we can get the real wave form by collecting the sampling data and it would not cause aliasing.

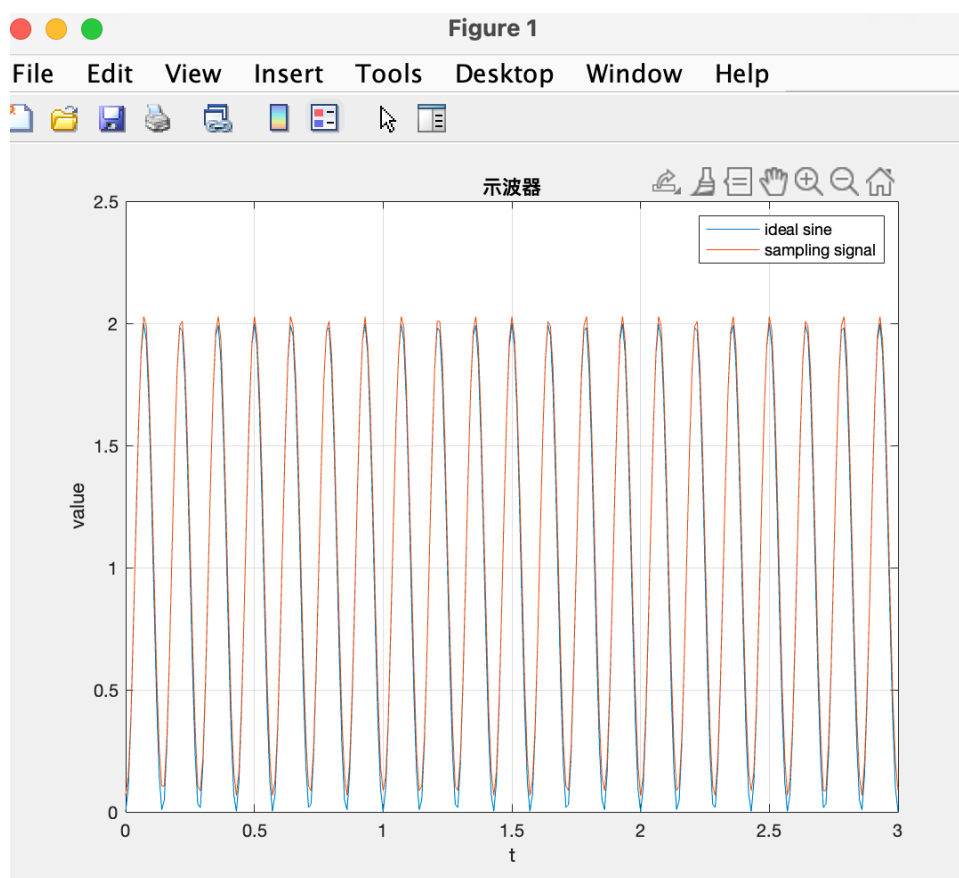
we can see from figure 1 that in time domain, for the 100Hz sampling rate, we did get the real wave form, which is a sine wave with $V_{pp}=2\text{V}$ offset=1V and frequency=5Hz. On the other hand, for sampling frequency=8.33Hz, the wave form plotted by sampling was not the wave form we put in.

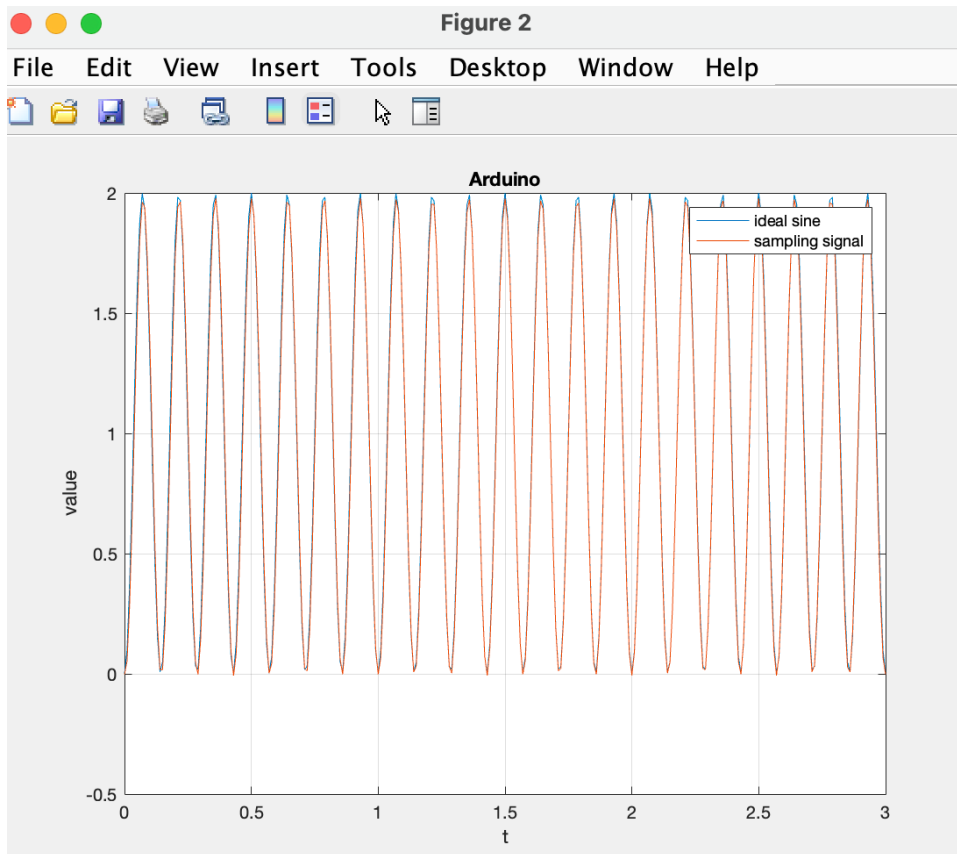
Besides, we can see from figure 2, from FFT,the frequency of the wave using sampling rate 100Hz is indeed 5Hz and from figure 3, with sampling rate=8.33Hz, the wave detected a false frequency as expected, which is about 3.33Hz.

Experiment 2: Quantization error

	Max error	Standard deviation
Theoretical Result	$<0.5\text{LSB}$	0.00564
Experiment Result(F.G)	+0.02/-0.12	
Experiment Result(Arduino)	0.11/-0.08	0.038

1. Result of the code plot(2 pics):





Discuss:

Is it what you want to get of this experiment result? Try to explain why we can not get the precise quantization error?

We can see from the above figure that the experiment result is quite reasonable, the standard deviation for arduino is slightly less than the data for function generator, which means that the gathered data is a more concentrated distribution, closer to the mean value. A low standard deviation doesn't necessary mean a more accurate data, however, in this case, we do know that the mean value=the accurate value, so a low standard deviation does mean more accuracy. Which does meet the given data, from the note, we know that $\text{max error} < \text{LSB}/2$ and arduino has a $\text{LSB}/2 = 2.5\text{mV}$, which is less than oscilloscope $= 9.8\text{mV}$, this means arduino does have a higher accuracy for ADC. And analog to digital means that we try to use digital values to model analog signal and the resolution determines the number of discrete levels available to represent the input range, however, it is still discrete, no matter how close the step, it is not continuous. And since we cannot get data less than the error gap, the quantization error we can have will be a range, not a precise value.

NOTE : Arduino: $\frac{\text{LSB}}{2} = \frac{5/1024}{2} = 0.0025\text{V} = 2.5\text{mV}$

$$\text{OSC: } \frac{LSB}{2} = \frac{5/256}{2} = 0.0098V = 9.8mV$$